Report: Configuring HTTP to HTTPS Redirection on an AWS Application Load Balancer (ALB)

Objective

The objective of this task was to configure an AWS Application Load Balancer (ALB) to automatically redirect all HTTP traffic to HTTPS, ensuring secure communication between users and the application.

Configuration Performed

The following configurations were implemented:

1. **Verified ALB Listeners**
    - Confirmed that the ALB had both:
        - HTTP Listener (Port 80)
        - HTTPS Listener (Port 443)
2. **Configured HTTP Redirect Rule**
    - Modified the HTTP (Port 80) listener to redirect all incoming requests to HTTPS.
    - The redirect rule was configured as follows:
        - **Protocol:** HTTPS
        - **Port:** 443
        - **Host:** `#{host}`
        - **Path:** `/#{path}`
        - **Query:** `#{query}`
        - **Status Code:** HTTP 301 (Permanent Redirect)
3. **Verified HTTPS Listener**
    - Confirmed that the HTTPS listener forwards traffic to the appropriate target group.
    - Verified that a valid ACM SSL/TLS certificate was attached to the HTTPS listener.
4. **Verified Security Group**
    - Confirmed that the ALB security group allows inbound traffic on:
        - TCP Port 80 (HTTP)
        - TCP Port 443 (HTTPS)

Issue Encountered

Although the redirect rule was correctly configured on the ALB, HTTP requests were not redirecting to HTTPS as expected.

Troubleshooting Performed

The following checks were carried out:

- Verified that both HTTP and HTTPS listeners exist.
- Confirmed that the redirect rule was correctly configured.
- Confirmed that the HTTPS listener forwards requests to the target group.
- Verified that the SSL certificate was successfully issued and attached.
- Confirmed that the ALB security group permits HTTP and HTTPS traffic.

Since the ALB configuration appeared correct, attention shifted to the DNS configuration.

Next Step

The next step is to verify the DNS configuration in **Amazon Route 53** by:

- Ensuring that the domain's **Alias A Record** points to the correct Application Load Balancer.
- Verifying that the domain is using the correct Route 53 **Name Servers** if the domain was registered with another registrar.
- Confirming that DNS propagation has completed.

Conclusion

The Application Load Balancer has been successfully configured to redirect HTTP traffic to HTTPS using a permanent (HTTP 301) redirect rule. Listener configuration, SSL certificate, and security group settings were verified and found to be correct. Since the redirect issue persists, the remaining area to investigate is the DNS configuration within Amazon Route 53 to ensure that the domain correctly resolves to the configured ALB.

AWS Cost Monitoring and Budget Alert Configuration Report

Objective

The objective of this exercise was to implement cost monitoring for an AWS environment by configuring AWS Budgets and Amazon Simple Notification Service (SNS). This ensures that spending is monitored proactively and that notifications are sent before costs exceed planned limits, particularly while working within the AWS Free Tier.

Activities Performed

1. Configured AWS Budget

- Accessed the **AWS Billing and Cost Management** console.
- Created a **Monthly Cost Budget** to monitor AWS spending.
- Defined an appropriate budget amount based on the expected Free Tier usage.
- Configured alert thresholds to notify when spending reaches predefined percentages of the budget (e.g., 80% and 100%).

2. Created an Amazon SNS Topic

- Opened the **Amazon SNS** console.
- Created a Standard SNS topic named **BudgetAlerts**.
- Added an email subscription to the topic.
- Confirmed the subscription through the email verification link.

3. Integrated AWS Budgets with SNS

- Linked the **BudgetAlerts** SNS topic to the AWS Budget notification settings.
- Configured the budget to publish notifications to the SNS topic whenever the defined thresholds are reached.

4. Notification Verification and Troubleshooting

After completing the configuration, email notifications were not received as expected. The following troubleshooting steps were performed:

- Verified that the SNS subscription had been confirmed.
- Confirmed that the correct SNS topic was attached to the AWS Budget.
- Reviewed the configured budget thresholds.
- Tested the SNS topic by publishing a test message.

- Checked spam and junk folders for the notification emails.

These checks were intended to determine whether the issue originated from the SNS configuration or the AWS Budget settings.

Expected Outcome

Once spending reaches the configured threshold, AWS Budgets publishes a notification to the **BudgetAlerts** SNS topic. Amazon SNS then forwards the notification to the subscribed email address, providing early warning of increasing AWS costs and helping prevent unexpected charges.

Conclusion

AWS Budgets and Amazon SNS were configured to provide automated cost monitoring and notifications. This implementation supports effective cloud cost management by enabling proactive monitoring of AWS usage and expenditure. During the process, notification delivery was validated through troubleshooting to ensure the alerting mechanism functions as intended.